

Global Technology: Reported HDD capacity additions by Toshiba: Framing implications for the HDD supply chain

What happened? STX and WDC traded 10%-11% lower on Friday following media reports of Toshiba's (NC) intention to increase HDD production capacity. Per a press release from its storage subsidiary, Toshiba is aiming to double its nearline HDD production capacity by the end of its FY27 (March 2028) vs FY25, with a ~¥60 bn (~\$380 mn) investment in its Philippines facility, according to Nikkei. Toshiba has also reportedly set a medium-term goal of tripling its overall HDD market share to 30% (from ~11% currently), and aims to achieve mass production of 65 TB drives in 2030. If achieved, at face value that would place Toshiba's technology roadmap in line with Seagate and Western Digital, which currently hold a multi-generation technology advantage.

Context and our view: Based on our recent checks with industry experts, we believe Toshiba does intend to expand its HDD production capacity - and this intention is validated by partner TDK's stated plans to expand head capacity. However, we would also note that at its 2022 Investor Day, Toshiba previously stated its intention to expand production, reach 25% market share by 2025 (from 17% at the time), and ramp 40TB capacity drives in the same timeframe - none of which came to fruition (vs. Toshiba's current share of 11%). However, given the HDD industry is both dramatically more profitable and growing faster today, we believe Toshiba is highly motivated to take advantage of these market conditions - and will likely attempt to execute on these plans.

Implications for STX (Buy) & WDC (Neutral): Even assigning a moderate probability of success for Toshiba in expanding capacity, we expect little change to the very healthy industry pricing and margin dynamics for both companies in the near term. We also see both companies - particularly STX - ahead in their HAMR technology roadmaps, which should allow them to sustain higher margins. However, in the medium term we do believe this dynamic could limit the degree of margin upside they can achieve over time. More important, we think this announcement could cap their equity multiples, especially in light of the very high premium (18X-20X NTM P/E) they currently enjoy over the memory complex (4X - 8X).

Implications for HDD components/materials supply chain:

We view this as rather having positive implications for the HDD components/materials supply chain. Unlike HDD manufacturers, most of the upstream HDD supply chain (centered on Japanese companies) has not implemented aggressive price hike policies, and their main benefits over the past 1-2 years have

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been an improved mix accompanying the increase in nearline HDDs and an increase in capacity utilization profits driven by higher volumes. Therefore, rather than concerns over intensified competition among suppliers or changes in market share triggered by Toshiba's potential production increase, we judge that positive factors such as a higher composition ratio of products with high technological difficulty and increased volumes across the entire industry will lead to higher expectations for earnings.

We inquired with TSR (Techno Systems Research), an industrial research expert company covering the HDD industry and others. Capacity share of the HDD market is determined by the number of units shipped and capacity per unit, and Toshiba's unit share is on an uptrend. This investment is presumed to be a move to address the delay in shifting to larger capacities (assumes this investment is for the MG13 series and later). On the other hand, considering the difficulty of launching production lines for large-capacity models in a short period of time, the view was also expressed that the hurdles to achieving goals for the technology roadmap, production capacity, and market share on a capacity basis are by no means low.

HDD-related component companies include: **(1) TDK** (HDD heads: mainly supplying Toshiba, but has also started shipping to STX/WDC. It is also a leading supplier of HDD suspensions); **(2) MinebeaMitsumi** (overwhelming market share in pivots. Supplies 100% of HDD-SPMs to STX); **(3) Nidec** (supplies 100% of HDD-SPMs to WDC/Toshiba); and **(4) Nitto Denko** (overwhelming market share in highly functional circuit materials).

In particular, **TDK (Buy)** holds a 100% share for HDD heads for Toshiba, and has been delivering/expanding shipments to STX and WDC since 2026 (we estimate this is mainly for the overflow of customers' in-house production capacity). HDD-related businesses account for just under 15% of TDK's company-wide operating profits (based on our FY3/27 estimate), but the profit margin is on a trajectory to rise sharply in tandem with the higher capacity utilization rate for HDD heads. In terms of the rate of change in earnings on a qoq/yoy basis, it is poised to contribute significantly to earnings expansion.

HDD-related materials companies include (1) **HOYA** (glass substrates) and (2) **Resonac** (selling HD media to Toshiba/WDC/STX).

In particular, **HOYA's (Buy)** glass substrates had previously only been adopted by STX for nearline HDDs, but sales contribution from a second company is expected to gradually begin from 2H FY3/27, and Toshiba also has a roadmap to adopt glass substrates starting with high-capacity models. HOYA has previously explained that the sales CAGR for glass substrates for 3.5-inch nearline HDDs is 10%+, but we think it has become highly likely that this outlook will be revised upward following this production capacity expansion by Toshiba (we assume an annual growth rate of 20% over FY3/28-FY3/29).

Also, we assume that Toshiba is the largest customer in **Resonac's** HD media business (the business accounts for the second-largest share of sales after copper clad laminates (CCL) in the semiconductor and electronic materials business, based on our FY12/26 estimates). Toshiba relies on external procurement for HD media. Based on data from TSR and other sources, assuming Toshiba's nearline shipments in 2026 are 10 mn units with 8-10 media disks per unit, we estimate that shipments for Toshiba will be 80 mn to 100 mn units, corresponding to 50%-60% of Resonac's current capacity of 160 mn units. Assuming that Toshiba's HDD capacity nearly doubles on a unit basis, Resonac's planned

capacity expansion of 50 mn units/year in Singapore, scheduled to sequentially come online in 2027, would likely be filled early by Toshiba alone. Consequently, we expect there would be little spare capacity for external sales to Seagate and Western Digital, which are also Resonac customers. We therefore believe supply/demand conditions could tighten considerably, increasing the likelihood of price hikes taking hold. In addition to the recently announced ([link](#)) 20 mn unit/year capacity expansion for nearline HD media for AI data centers (which will cumulatively increase capacity by c.44% from the current c.160 mn units/year to a total of c.230 mn units/year from 2029), further capacity expansions could come into view going forward. Including the price hike initiatives that Resonac has been rolling out in 2H12/26, we view this news as increasing our conviction in the medium- to long-term growth potential of the HD media business, a core operation within the semiconductor and electronic materials business.

Valuation and Risks:

Seagate (Buy, covered by James Schneider): We are Buy rated on STX. Our 12-month price target of \$960 is based on a 24X P/E multiple applied to our normalized EPS estimate of \$40.00. Key downside risks: (1) difficulties qualifying HAMR at the major hyperscalers; (2) lack of supply-side prudence from Seagate or the broader industry.

Western Digital (Neutral, covered by James Schneider): We are Neutral rated on WDC. Our 12-month price target of \$615 is based on a 22X P/E multiple applied to our normalized EPS of \$28.00. Key upside/downside risks: (1) any pull-in of the company's HAMR roadmap; (2) new consumer/enterprise storage applications; (3) multiple compression due to industry overshipping end-demand; (4) any signaling of HAMR roadmap push outs or incremental technical challenges.

TDK (Buy, covered by Daiki Takayama): We are Buy rated on TDK with a 12-month price target of ¥4,900. Our target price is based on a FY3/29E EV/GCI vs. CROCI/WACC, applying a 10% premium to our sector average EV/DACF multiple of 10X (implies FY3/28E P/E of 27X). Key risks: Decline in smartphone production volume, higher input costs, and yen appreciation.

HOYA (Buy, covered by Shuhei Nakamura): Our 12-month target price of ¥36,000 is based on a sum-of-the-parts valuation methodology (FY28E EV/EBITDA-based, implies FY3/28E P/E of 36X and P/B of 12X), using different multiples for each of HOYA's two main segments – life care (FY3/28E EV/EBITDA of 18X) and IT (FY3/28E EV/EBITDA of 28X). Key risks: Intensifying competition in core businesses, a weakening of demand, and large-scale capex or M&A deals.

Resonac (Buy, covered by Atsushi Ikeda): We are Buy rated on Resonac with a 12-month target price of ¥25,060. Our target price is based on the correlation between the materials sector's EV/GCI and the FY12/27E CROCI/WACC (cash-return multiple of 0.7X plus a 30% premium to the sector average). Key risks include a slowdown in semiconductor demand, yen appreciation, and decline in graphite electrode earnings.

Disclosure Appendix

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We, James Schneider, Ph.D., Daiki Takayama, Shuhei Nakamura, Atsushi Ikeda, Khalil Fenina, Kaho Otake, Yuri Izumikawa, Makoto Takahara, Yuji Hidaka, Marshall Wong, Anmol Makkar, Luya You and Liana Wadhvani, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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Goldman Sachs is a registered market maker in the securities or derivatives thereof: Seagate Technology (\$848.99) and Western Digital Corp. (\$415.29)

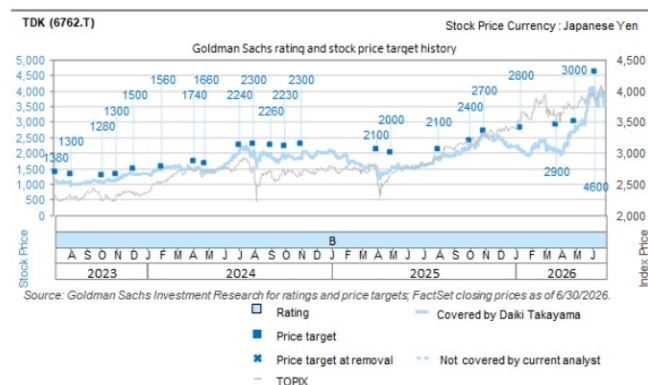
Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

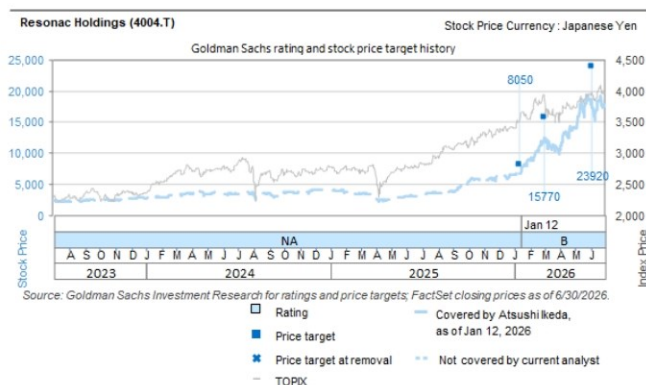
	Rating Distribution				Investment Banking Relationships		
	Buy	Hold	Sell		Buy	Hold	Sell
Global	50%	34%	16%		65%	59%	43%

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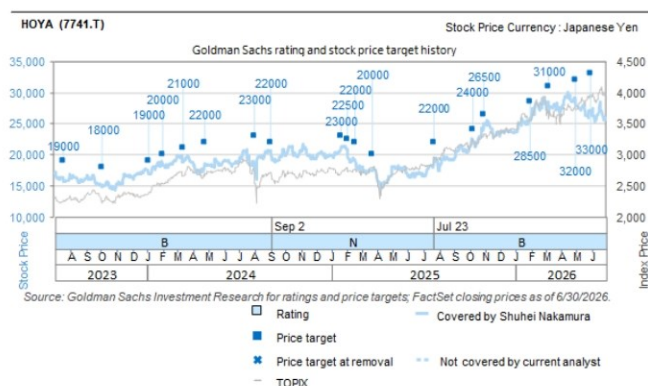
Price target and rating history chart(s)



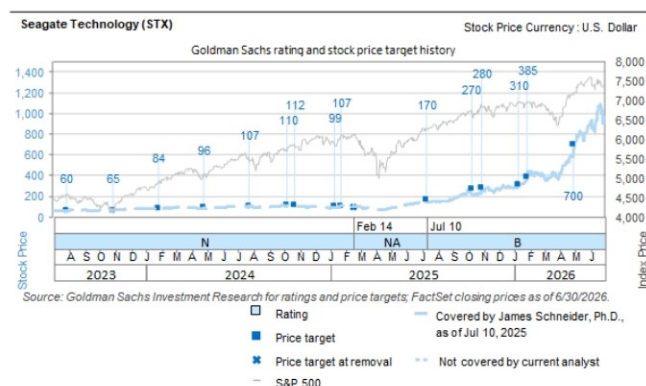
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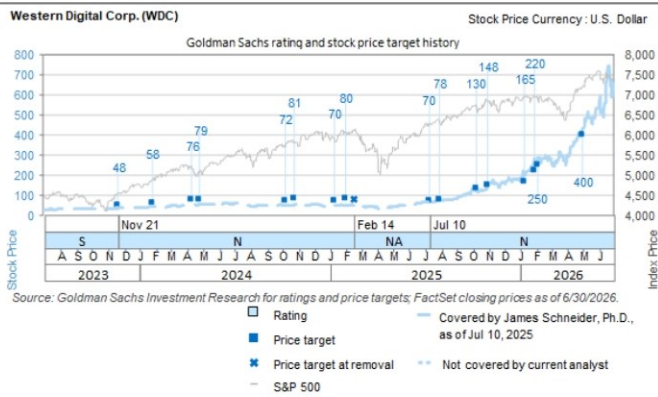
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Target price history table(s)

Seagate Technology (STX)

Date of report	Target price (\$)	Closing price (\$)
05-Jul-26	960.00	820.16
29-Apr-26	700.00	643.30
28-Jan-26	385.00	442.93
09-Jan-26	310.00	304.01
28-Oct-25	280.00	223.00
08-Oct-25	270.00	224.35
10-Jul-25	170.00	144.50
22-Jan-25	107.00	108.18
10-Jan-25	99.00	88.87
23-Oct-24	112.00	103.52
09-Oct-24	110.00	109.29
24-Jul-24	107.00	109.53
24-Apr-24	96.00	87.11
25-Jan-24	84.00	90.99
27-Oct-23	65.00	68.20

Western Digital Corp. (WDC)

Date of report	Target price (\$)	Closing price (\$)
06-Aug-26	615.00	451.52
05-Jul-26	650.00	539.00
01-May-26	400.00	431.52
03-Feb-26	250.00	290.24
30-Jan-26	220.00	250.23
09-Jan-26	165.00	200.46
30-Oct-25	148.00	138.13
08-Oct-25	130.00	121.18
31-Jul-25	78.00	78.69
10-Jul-25	70.00	65.06
30-Jan-25	80.00	49.71
10-Jan-25	70.00	47.47
25-Oct-24	81.00	52.48
09-Oct-24	72.00	48.72
26-Apr-24	79.00	53.92
12-Apr-24	76.00	54.45
26-Jan-24	58.00	44.00
21-Nov-23	48.00	34.94

HOYA (7741.T)

Date of report	Target price (¥)	Closing price (¥)
23-Aug-26	36,000	24,895
31-Jul-26	35,000	24,675
31-May-26	33,000	27,080
30-Apr-26	32,000	29,100
09-Mar-26	31,000	26,765
30-Jan-26	28,500	25,870
31-Oct-25	26,500	25,085
08-Oct-25	24,000	22,155
23-Jul-25	22,000	18,590
24-Mar-25	20,000	17,490
18-Feb-25	22,000	19,040
03-Feb-25	22,500	19,625
22-Jan-25	23,000	21,500
02-Sep-24	22,000	20,760
01-Aug-24	23,000	18,680
24-Apr-24	22,000	18,685
13-Mar-24	21,000	18,925
01-Feb-24	20,000	18,550
04-Jan-24	19,000	16,995

Resonac Holdings (4004.T)

Date of report	Target price (¥)	Closing price (¥)
31-Aug-26	25,060	15,610
05-Jun-26	23,920	17,058
01-Mar-26	15,770	11,753
12-Jan-26	8,050	6,709

HOYA (7741.T)

Date of report	Target price (¥)	Closing price (¥)
05-Oct-23	18,000	-

Resonac Holdings (4004.T)

Date of report	Target price (¥)	Closing price (¥)
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TDK (6762.T)

Date of report	Target price (¥)	Closing price (¥)
31-Jul-26	4,900	3,008
08-Jun-26	4,600	3,715
28-Apr-26	3,000	2,677
23-Mar-26	2,900	2,043
12-Jan-26	2,800	2,142
31-Oct-25	2,700	2,673
02-Oct-25	2,400	2,154
01-Aug-25	2,100	1,876
28-Apr-25	2,000	1,460
31-Mar-25	2,100	1,546
01-Nov-24	2,300	1,848
01-Oct-24	2,230	1,948
02-Sep-24	11,300	2,023
30-Jul-24	11,500	2,029
02-Jul-24	11,200	2,004
26-Apr-24	8,300	1,462
03-Apr-24	8,700	1,488
31-Jan-24	7,800	1,488
06-Dec-23	7,500	1,338
01-Nov-23	6,500	1,158

Price targets shown in table(s) are unadjusted for corporate actions.

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